CENTER FOR DRUG EVALUATION AND RESEARCH

APPLICATION NUMBER:

211855Orig1s000

CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS REVIEW(S)

OFFICE OF CLINICAL PHARMACOLOGY REVIEW

NDA or BLA Number	211855
Submission Date	Dec 13, 2018
Submission Type	505(b)(2) NDA
Brand Name	Vumerity
Generic Name	Diroximel fumarate (ALKS 8700)
Dosage Form and Strength	delayed release capsules, 231 mg
Route of Administration	oral
Proposed Indication	Treatment of Relapsing forms of Multiple Sclerosis (RMS)
Applicant	Alkermes
OCP Review Team	Hristina Dimova, Ph.D., Angela Men, M.D., Ph.D.

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1. EXECUTIVE SUMMARY

Diroximel fumarate (DRF, ALKS 8700) is 2-succinimidolethyl methyl fumarate, an aminoethyl ester of monomethyl fumarate (MMF) that undergoes rapid presystemic hydrolysis by esterases to produce MMF (active form of dimethyl fumarate (DMF)). The sponsor is seeking approval for Vumerity (Diroximel fumarate) by submitting a 505(b)(2) application with reliance on the FDA's previous finding of the safety and effectiveness of DMF for the treatment of relapsing forms of MS (Tecfidera; NDA 204063).

As previously agreed with the FDA, the main efficacy evaluation for DRF is based on a pharmacokinetic (PK) bridging approach to the listed drug (LD), Tecfidera (DMF delayed-release capsules, 240 mg). Since MMF is the active metabolite of both DRF and DMF, the bridging approach was to demonstrate comparative MMF PK data for DRF and DMF. This PK approach was further supported by ten clinical pharmacology studies to qualify and characterize DRF and its metabolites, e.g. characterize absorption, distribution, metabolism, and excretion, effects on intrinsic and extrinsic factors, and effects on QT interval. The two ongoing Phase 3 studies in subjects with RMS (Studies A301 and A302) were designed as long-term safety studies.

Diroximel fumarate 462 mg met the bioequivalence (BE) criteria for both AUC and Cmax parameters of MMF relative to DMF 240 mg, under fasted conditions. DRF administered with a high-fat/high/calorie meal met the criteria for BE to DMF based on AUC for MMF, however mean Cmax for MMF was 26% lower compared to that of DMF administered under the same (high fat) conditions, which is out of BE criteria. Administration of DRF after either a low-fat/low-calorie or a medium-fat/medium-calorie meal resulted in MMF exposure within the established MMF Cmax window for DMF, establishing the PK bridge between DRF and DMF under these dietary conditions.

A consult was sent to the Office of Scientific Inspections and Surveillance (OSIS) requesting clinical and bioanalytical site inspections for BE studies ALK8700-A103, ALK8700-A104 and ALK8700-A109. OSIS concluded that the data are acceptable based on the records of recent inspections for two of the clinical and bioanalytical sites, Alkermes, Inc. and (b) (4) OSIS conducted an inspection of (b) (4) and concluded that the data from the audited studies are reliable to support a regulatory decision (Report issued July 26, 2019).

1.1 Recommendations

The office of Clinical Pharmacology (OCP) has reviewed the information contained in NDA 211855 and finds it acceptable from an OCP perspective. Key review issues with specific recommendations and comments are summarized below:

Review Issue	Recommendations and Comments
Pivotal or supportive evidence of effectiveness	Diroximel fumarate 462 mg met the BE criteria for both AUC and Cmax parameters of MMF relative to DMF 240 mg, under fasted conditions. Administration of DRF after either a low-fat/low-calorie or a medium-fat/medium-calorie meal resulted in MMF exposure within the established MMF Cmax window for DMF. DRF administered with a high-fat/high/calorie meal met the criteria for BE to DMF based on AUC for MMF, however mean Cmax for MMF was 26% lower compared to that of DMF.
General dosing instructions	The dosing recommendations are based on LD and the established PK bridging. Initial titration with a starting dose of 231 mg Vumerity twice a day (bid) for 7 days is recommended, the dose should be increased to the maintenance dose of 462 mg Vumerity bid. Administration of Vumerity with a high-fat, high- calorie meal and with alcohol should be avoided.
Dosing in patient subgroups (intrinsic and extrinsic factors)	Not recommended in patients with moderate or severe renal impairment due to significant increase in the exposure of DRF major metabolite (HES).
Labeling	General dosing recommendations are acceptable. Administration of Vumerity with a high-fat, high- calorie meal and with alcohol should be avoided.
Bridge between the to-be-marketed and clinical trial formulations	The to-be-marketed formulation was used in the pivotal BE studies.

2. SUMMARY OF CLINICAL PHARMACOLOGY ASSESSMENT

2.1 Pharmacology and Clinical Pharmacokinetics

Dimethyl fumarate, a pro-drug of monomethyl fumarate, is indicated for the treatment of relapsing forms of MS. Similar to DMF, DRF undergoes rapid pre-systemic hydrolysis by esterases and is converted to the active metabolite, MMF. The mechanism by which MMF exerts its therapeutic effect in multiple sclerosis is unknown. Monomethyl fumarate has been shown to activate the Nuclear factor (erythroid-derived 2)-like 2 (Nrf2) pathway in vitro and in animals and humans. The Nrf2 pathway is involved in the cellular response to oxidative stress. MMF has been identified as a nicotinic acid receptor agonist in vitro.

2.1.1 Pharmacokinetic Comparison between Diroximel fumarate and Dimethyl fumarate

Diroximel fumarate 462 mg met the bioequivalence (BE) criteria for both AUC and Cmax parameters of MMF relative to DMF 240 mg, under fasted conditions. DRF administered with a high-fat/high/calorie meal met the criteria for BE to DMF based on AUC for MMF, however mean Cmax for MMF was 26% lower compared to that of DMF administered under the same (high fat) conditions. Administration of DRF after either a low-fat/low-calorie or a medium-fat/medium-calorie meal resulted in MMF exposure within the established MMF C_{max} window for DMF, see Section 3.3 for further details.

Note: All clinical studies were conducted using the proposed delayed release commercial formulation, with the exception of Study A105 (that used radiolabeled DRF).

2.1.2 Effect of Food on the Pharmacokinetics of Monomethyl Fumarate

Following oral administration of DRF, the T_{max} of MMF was delayed from 2.5 hours (fasted state) to 4.5 h (low-fat, low-calorie meal or a medium-fat, medium-calorie meal) and 7.0 hours (high fat, high calorie meal). A high-fat, high-calorie meal (900-1000 calories, 50% of calories from fat) resulted in a 41-44% reduction in MMF mean C_{max} compared to fasted state. The MMF C_{max} with low-fat, low-calorie (350 to 400 calories, 10 to 15g fat) and medium-fat, medium-calorie (650 to 700 calories, 25-30g fat) meals was reduced by approximately 12% and 25%, respectively. There was no impact of low, medium, or high-fat meals on the AUC of MMF after administration of DRF.

2.2 Dosing and Therapeutic Individualization

2.2.1 General dosing

The dosing recommendations are based on LD and the established PK bridging. Initial titration with a starting dose of 231 mg Vumerity twice a day (bid) for 7 days is recommended, the dose should be increased to the maintenance dose of 462 mg Vumerity (administered as two 231 mg capsules) bid. Administration of Vumerity with a high-fat, high- calorie meal should be avoided.

Administration of Vumerity with 5% v/v and 40% v/v ethanol did not induce dose dumping, however the MMF C_{max} was significantly reduced for the 40% v/v ethanol solution (more than 50% in 10 out 29 subjects). Vumerity should not be used with alcohol.

2.2.2 Therapeutic individualization

The plasma exposures of the major metabolite of DRF, HES, were significantly increased in subjects with moderate and severe renal impairment relative to healthy subjects. The impact of such HES exposure increase on the safety profile is unknown. VUMERITY is not recommended in patients with moderate or severe renal impairment.

2.3 Post-Marketing Commitment

None is identified.

2.4 Summary of Labeling Recommendations

Sponsor proposed language in Dosing and Administration that VUMERITY (b) (4) (b) (4) a high-fat, high- calorie meal should be avoided. <u>VUMERITY</u> (b) (4) (b) (4) could be confusing for the patient. <u>Avoid high-fat, high- calorie meal</u> is acceptable.

Administration of Vumerity with alcohol should be avoided.

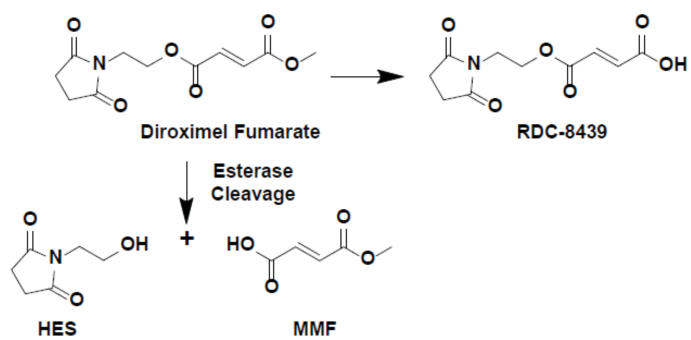
Vumerity is not recommended in patients with moderate or severe renal impairment.

3. COMPREHENSIVE CLINICAL PHARMACOLOGY REVIEW

3.1 Overview of the Product and Regulatory Background

This is a 505(b)(2) application with reliance on the FDA's previous finding of the safety and effectiveness of DMF for the treatment of relapsing forms of MS (Tecfidera; NDA 204063). Diroximel fumarate is an aminoethyl ester of monomethyl fumarate (MMF) that undergoes hydrolysis to produce MMF (the active form of DMF). Additionally, a major metabolite (HES) and a minor metabolite (RDC-8439), not present after DMF administration, are produced.

Figure 1: Conversion of diroximel fumarate to Monomethyl Fumarate (MMF)



At the EOP2 meeting (Sep 2015) the Division stated that PK data may be sufficient to provide a bridge to efficacy data for Tecfidera assuming that comparative bioavailability studies provide adequate bridge to the reference listed drug and that the major metabolite RDC-6567 (HES) is adequately evaluated.

In a follow-up Type C interaction (May 2016), the Agency advised that the MMF exposure (C_{max}) with food for DRF should not be lower than the reduced MMF exposure observed for Tecfidera with food. (Taking Tecfidera with a high-fat, high-calorie meal did not affect the AUC of MMF but decreased its C_{max} by 40%). There was a concern that a reduced MMF C_{max} after DRF compared to that after 240 mg Tecfidera with food may affect the efficacy of DRF, since exposure-response relationship for safety and efficacy is not available for MMF.

3.2 General Pharmacology and Pharmacokinetic Characteristics

Pharmacology	
Mechanism of Action	DRF undergoes rapid pre-systemic hydrolysis and is converted to the active metabolite, MMF. The mechanism by which MMF exerts its therapeutic effect in multiple sclerosis is unknown. Monomethyl fumarate has been shown to activate the Nuclear factor (erythroid-derived 2)-like 2 (Nrf2)

	pathway in vitro and in vivo in animals and humans. The Nrf2 pathway is involved in the cellular response to oxidative stress.
General Information	
Bioanalysis	Concentrations of DRF and its metabolites MMF, HES, and RDC-8439 were measured by validated bioanalytical methods in plasma. The bioanalytical methods utilized liquid chromatography with tandem mass spectrometry (LC-MS/MS) detection and were developed and validated following regulatory guidelines by the bioanalytical laboratories of [redacted] (b) (4)
Healthy Volunteers vs Patients	NA The BE studies were conducted in healthy subjects.
ADME	
Absorption	Diroximel fumarate undergoes rapid pre-systemic hydrolysis and is converted to the active metabolite, MMF. DRF is not quantifiable in plasma following oral administration of Vumerity. The median Tmax of MMF is 2.5 to 3 hours in the fasted state. After low-fat, low-calorie meal or a medium-fat, medium-calorie meal, MMF Tmax was 4.5 h and after high fat, high calorie meal MMF Tmax was 7.0 hours. There was no impact of low, medium, or high-fat meals on the AUC of MMF after administration of Vumerity.
Distribution	The apparent volume of distribution for MMF is between 72 L and 83 L in healthy subjects after administration of Vumerity. Human plasma protein binding of MMF was 25%.
Metabolism	Diroximel fumarate is extensively metabolized by esterases before it reaches the systemic circulation. Further metabolism of MMF occurs through the tricarboxylic acid (TCA) cycle, with no involvement of the cytochrome P450 system [Tecfidera]. Fumaric and citric acid, and glucose are the major metabolites of MMF in plasma. Esterase metabolism of DRF produces MMF, the active metabolite, and 2-hydroxyethyl succinimide (HES), an inactive major metabolite.
Elimination	MMF is mainly eliminated as carbon dioxide in the expired air with only trace amounts recovered in urine [Tecfidera]. The terminal half-life of MMF is approximately 1 hour. HES is mainly eliminated in urine with terminal half-life between 10 and 15 hours.

3.3 Clinical Pharmacology Review Questions

3.3.1 To what extent does the available clinical pharmacology information provide pivotal or supportive evidence of effectiveness?

No efficacy studies were conducted for this NDA. The pivotal evidence of effectiveness relies on the PK bridging between Vumerity and the listed drug, DMF (Tecfidera; NDA 204063). The sponsor conducted three BE/ Relative Bioavailability and Food Effect studies (i.e., Studies A103, A104, and A109) to compare DRF and DMF with regard to the exposures of the active metabolite, MMF, under fasted and fed states with varying conditions of fat and caloric content. Two long-term Phase 3 safety studies in subjects with RMS are ongoing.

Study A103 was a single-dose, randomized, two-sequence, two-treatment, crossover, double-blind study to evaluate the PK and safety of DRF and DMF in healthy subjects when administered in the fasted state. Study drug was administered on Day 1 and Day 8 after fasting for a minimum of 10 hours pre-dose. Subjects continued to fast until 4 hours post-dose. Concentrations of DRF and its metabolites (MMF, HES, and RDC-8439) were quantified in plasma samples.

A total of 35 subjects were randomized in the study; 18 subjects received DRF 462 mg (2 capsules of 231 mg) followed by DMF 240 mg (1 capsule of 240 mg and 1 capsule of placebo) in Sequence 1, and 17 subjects received DMF 240 mg followed by DRF 462 mg in Sequence 2.

Study A103 summary results:

• MMF and HES were the primary plasma metabolites of DRF. The mean (SD) metabolite ratio, calculated as the AUC_{last} of HES to MMF, was 31.8 (8.28).

• Diroximel fumarate was not detectable in plasma, and RDC-8439 concentrations were not measurable (except in 2 subjects) following administration of DRF.

• Diroximel fumarate met the criteria for BE to DMF with regard to the exposure for MMF (both Cmax and AUC), when administered in the fasted state.

Summary of ANCOVA Results for Plasma MMF AUC_{last}, AUC_{∞}, and C_{max} Following a Single Oral Administration of Diroximel Fumarate 462 mg (Test) Versus Dimethyl Fumarate 240 mg (Reference) Formulations in the Fasted State (Study A103)

Parameter[a]	Geometric Means (SE) n		Geometric Mean Ratio (%)	90% CI Lower, Upper
	DRF 462 mg (N=35)	DMF 240 mg (N=35)		
C_{max} (μg/mL)	1.57 (0.07) 35	1.67 (0.07) 35	0.94	0.82, 1.07
AUC_{last} (μg·hr/mL)	3.38 (0.05) 35	3.14 (0.05) 35	1.08	1.00, 1.16
AUC_{∞} (μg·hr/mL)	3.57 (0.06) 22	3.56 (0.07) 14	1.00	0.88, 1.14

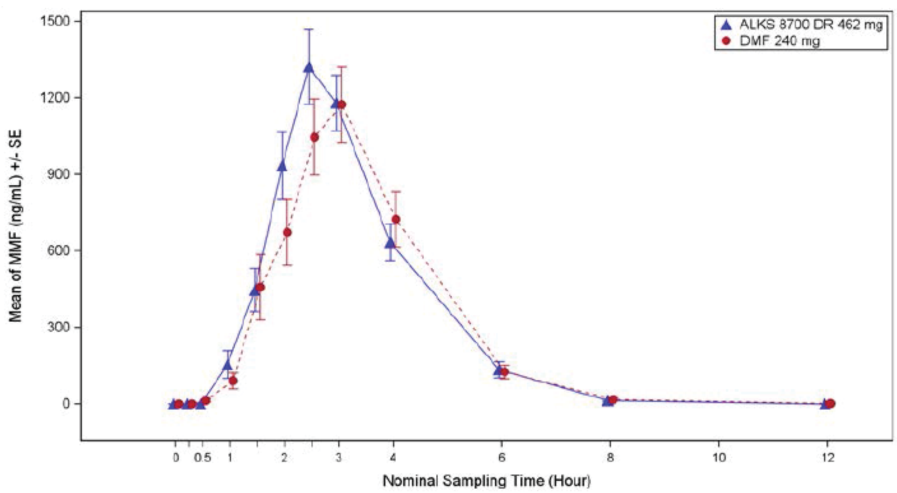
Abbreviations: ANCOVA=analysis of covariance; AUC_{∞}=area under the concentration-time curve from time zero extrapolated to infinity; AUC_{last}=area under the concentration-time curve from time zero until the last measurable concentration time point; CI=confidence interval; C_{max}=maximum plasma concentration; DMF=dimethyl fumarate; DRF=diroximel fumarate; MMF=monomethyl fumarate; N=number of subjects who received a specific treatment (subjects could be included in more than one treatment group); n=number of subjects included in the calculation of the geometric least squares means and SE, which is the number of subjects with specific PK data while on any of the dose regimens; PK=pharmacokinetic; SE=standard error.

[a] Based on an ANCOVA model, including treatment sequence, treatment group, and period as fixed factors and subject within treatment sequence as a random factor.

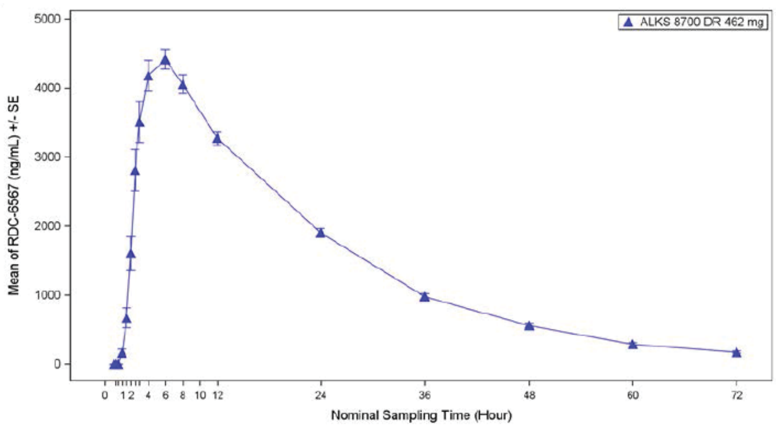
Note: The SE displayed is in logarithmic scale.

Source: Module 5.3.1.2 Study A103, Table 7.

(Mean±SE) MMF Plasma Concentration after Administration of DRF and DMF in the Fasted State (PK Population; Study A103)



Plasma Concentration (Mean±SE) of HES (PK Population; Study A103)



Studies A104 and A109 were conducted to compare DRF and DMF with regard to the exposures of MMF, under fed states with varying conditions of fat and caloric content, see Section 3.3.4 for details.

3.3.2 Is the proposed dosing regimen appropriate for the general patient population for which the indication is being sought?

Yes, the proposed dosing regimen is acceptable. The dosing recommendations for Vumerity are based on these of the LD (Tecfidera) and the established PK bridge.

Refer to Sections 3.2.1 and 3.3.4 for details.

3.3.3 Is an alternative dosing regimen and/or management strategy required for subpopulations based on intrinsic factors?

Based on the results from the human mass-balance study (ALK8700-A105), RDC-6567 is a major metabolite of DRF and is predominantly excreted in urine. Therefore, a clinical study (A108) was designed to assess effects of renal impairment on the PK of MMF (the main therapeutic moiety) and RDC-6567 (the major metabolite of DRF) following a single dose administration of DRF.

Subjects in the mild, moderate, and severe renal impairment cohorts were assigned using the eGFR ranges of 60 to 89, 30 to 59, and <30 mL/min/1.73 m^2, respectively. Subjects were required to fast overnight (for at least 8 hours). Subjects were provided a standardized light snack 2 hours prior to dosing, and water consumption was restricted from 1 hour prior to dosing until 2 hours after dosing. All subjects received a single oral dose of DRF (462 mg provided as two 231-mg capsules) with 240 mL of water. No food was allowed for at least 4 hours after dosing.

PK analysis included plasma concentrations of DRF, MMF, HES, and RDC-8439 and urine concentrations of MMF and HES.

Overall, PK values of MMF exhibited a high degree of variability within each cohort but were generally similar across cohorts (per the sponsor).

Comments: Sponsor claims that the range of MMF values in each cohort overlapped between cohorts, and no trends with respect to severity of renal impairment were observed, however MMF exposure was increased in subjects with renal impairment, including mild, see tables below (especially the max range increased from 4.4 to 8.3 h•µg/mL).

Study ALK8700-A108: MMF Plasma Pharmacokinetic Parameters by Cohort

Parameter Statistics	Renal Function Cohort			
	Healthy (N=8)	Mild Impairment (N=8)	Moderate Impairment (N=8)	Severe Impairment (N=8)
AUC_∞ (h•µg/mL)				
n	5	5	6	6
Mean (SD)	3.336 (1.098)	4.879 (1.884)	4.939 (2.343)	4.743 (2.154)
Geometric mean (CV%)	3.179 (32.9)	4.544 (38.6)	4.430 (47.4)	4.348 (45.4)

MMF

Parameter Statistics	Renal Function Cohort			
	Healthy (N=8)	Mild Impairment (N=8)	Moderate Impairment (N=8)	Severe Impairment (N=8)
AUC_∞ (h*ug/mL)				
n	5	5	6	6
Mean (SD)	3.336 (1.098)	4.879 (1.884)	4.939 (2.343)	4.743 (2.154)
CV (%)	32.9	38.6	47.4	45.4
Median	3.410	4.426	4.915	4.636
Min - Max	1.929 - 4.399	2.281 - 7.007	1.894 - 8.457	2.306 - 8.284
Geometric Mean	3.179	4.544	4.430	4.348
AUC_{last} (h*ug/mL)				
n	8	8	8	8
Mean (SD)	2.843 (1.233)	4.450 (1.723)	4.945 (2.099)	4.426 (2.116)
CV (%)	43.4	38.7	42.4	47.8
Median	2.930	4.335	4.875	4.520
Min - Max	0.962 - 4.370	2.260 - 6.990	1.870 - 8.440	1.810 - 8.270
Geometric Mean	2.566	4.142	4.515	3.976

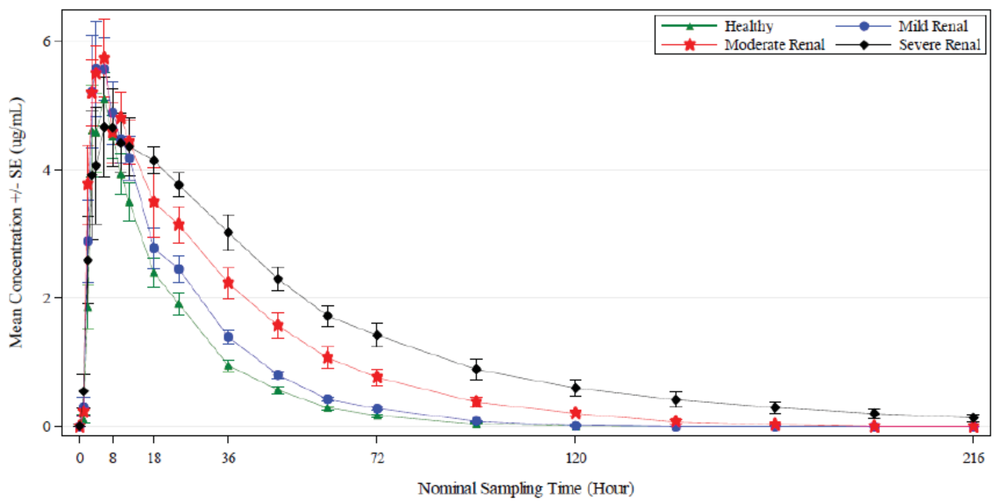
Renal clearance of MMF was decreased as well in moderate/severe impaired groups:

Summary of Renal Clearance in Urine of Monomethyl Fumarate by Cohort

Parameter Statistic	Renal Function Cohort			
	Healthy (N=8)	Mild Impairment (N=8)	Moderate Impairment (N=8)	Severe Impairment (N=8)
CL_R (L/h)				
n	8	8	8	8
Mean (SD)	0.097 (0.028)	0.126 (0.073)	0.046 (0.024)	0.050 (0.050)
CV (%)	28.3	58.1	51.2	100.3
Median	0.102	0.116	0.051	0.029
Min-Max	0.047 - 0.138	0.037 - 0.278	0.007 - 0.073	0.001 - 0.129
Geometric Mean	0.093	0.109	0.037	0.022

HES (RDC-6567): The C_{max} and t_{max} of HES in all cohorts were similar, while AUC increased with increasing renal impairment. HES was undetectable in plasma after 120 hours in the healthy and mild impairment cohorts, whereas measurable concentrations persisted for 216 hours in the severe impairment cohort. An increase was observed in the mean AUC_{∞} for HES, which increased from a mean (SD) of 111 (26.5) µg·hr/mL in the healthy cohort to 295 (80.1) µg·hr/mL in the severe impairment cohort.

ALK8700-A108: HES Plasma Concentration (Mean±SE)



CSR=Clinical Study Report; HES=2-hydroxyethyl succinimide; LLOQ=lower limit of quantification; PK=pharmacokinetic; SE=standard error.
Values below the LLOQ were set to 0 in the calculation of the summary statistics.
Source: CSR A108 Figure 4.

Study ALK8700-A108: Summary of Plasma PK Parameters for HES (RDC-6567) by Cohort

Parameter Statistics	Renal Function Cohort			
	Healthy (N=8)	Mild Impairment (N=8)	Moderate Impairment (N=8)	Severe Impairment (N=8)
AUC_∞ (h•µg/mL)				
n	8	8	8	8
Mean (SD)	111.146 (26.487)	138.971 (28.851)	203.505 (64.984)	295.025 (80.143)
Geometric mean (CV%)	108.938 (23.8)	136.274 (20.8)	193.581 (31.9)	285.718 (27.2)

Comments: Moderate or severe renal impairment were excluded in the phase 3 safety studies ALK8700-A301 and ALK8700-A302. However, six patients had eGFR <= 60 in study A301 were identified. There are very limited safety data in these patient populations. In addition, in the nonclinical studies, HES exposures in that range generally showed toxicity. The 3x increase in HES exposure would be over the NOAELs.

Of the 695 subjects in ALK8700-A301, 170 had a baseline GFR < 90 mL/min/1.73m^2. Of the 59 subjects randomized to DRF in ALK8700-A302, 20 had a baseline GFR < 90 mL/min/1.73m^2. There were no safety concerns in the mild renal impairment subpopulation attributable to the study drug.

Recommendations: No dosing adjustment is recommended patients with mild renal impairment. Vumerity is not recommended in moderate and severe renal impairment.

3.3.4 Are there clinically relevant food-drug interactions and what is the appropriate management strategy?

Studies A104, and A109 were conducted to compare DRF and DMF with regard to the exposures of the active metabolite, MMF, under fasted and fed states with varying conditions of fat and caloric content.

Study A104 was a single-center, randomized, two-sequence, two-treatment, crossover, double-blind study to evaluate the comparative bioavailability (BA) of MMF following administration of 462 mg DRF (2 capsules of 231 mg) to 240 mg DMF (1 capsule of 240 mg and 1 capsule of placebo), in healthy subjects in fed conditions. Subjects consumed a high-fat (approximately 50% of total caloric content of the meal) and high-calorie (approximately 900 to 1000 calories) morning meal during the 30 minutes before the study drug was administered.

A total of 42 subjects were randomized in the study; 21 subjects received DRF 462 mg followed by DMF 240 mg in Sequence 1, and 21 subjects received DMF 240 mg followed by DRF 462 mg in Sequence 2.

Study 104 PK Summary: The MMF PK profiles for both DRF and DMF administered with a high-fat and high-calorie meal were characterized by multiple peaks and a median t_{lag} of 3.5 and 4 hours for DRF and DMF.

For MMF AUC_{last} and AUC_{∞}, the geometric mean ratios were close to 1 and the 90% CIs were contained completely within the standard boundaries of 0.8 to 1.25 used to evaluate BE.

The MMF mean C_{max} was 26% lower for DRF as compared with DMF under fed conditions.

The mean (SD) metabolite ratio, calculated as the AUC_{last} of HES to MMF, was 31.6 (8.63).

Summary of ANCOVA Results for Plasma MMF AUC_{last}, AUC_{∞}, and C_{max} Following a Single Oral Administration of DRF 462 mg (Test) Versus DMF 240 mg (Reference) Formulations with a High-Fat and High-Calorie Meal (Study A104)

Parameter[a]	Geometric Means (SE) n		Geometric Mean Ratio (%)	90% CI Lower, Upper
	DRF 462 mg (N=42)	DMF 240 mg (N=42)		
C_{max} (µg/mL)	0.88 (0.08) 42	1.18 (0.08) 42	0.74	0.64, 0.85
AUC_{last} (µg·hr/mL)	2.69 (0.04) 42	2.80 (0.04) 42	0.96	0.89, 1.03
AUC_{∞} (µg·hr/mL)	2.83 (0.05) 25	3.08 (0.05) 25	0.92	0.84, 1.01

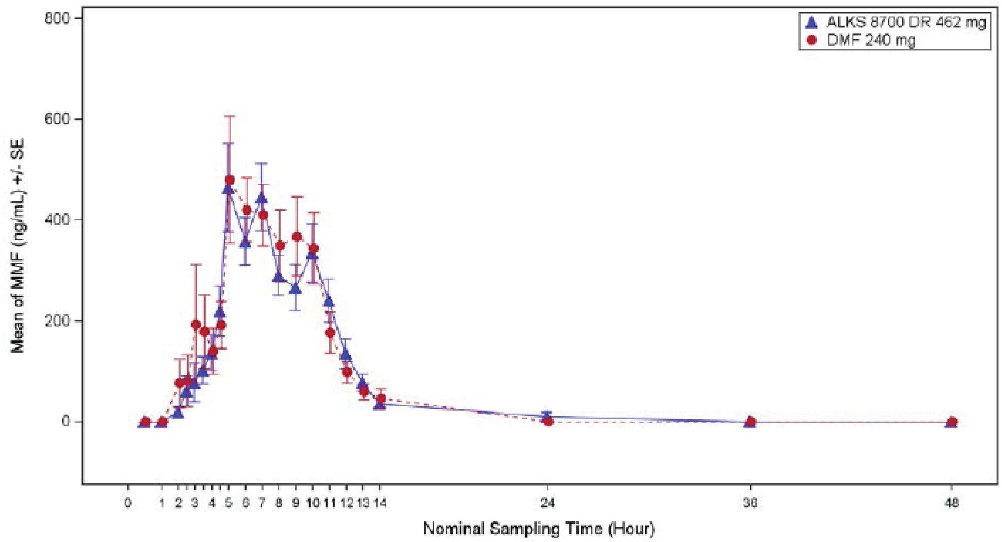
Abbreviations: ANCOVA=analysis of covariance; AUC_{∞}=area under the concentration-time curve from time zero extrapolated to infinity; AUC_{last}=area under the concentration-time curve from time zero until the last measurable concentration time point; CI=confidence interval; C_{max}=maximum plasma concentration; DMF=dimethyl fumarate; DRF=diroximel fumarate; MMF=monomethyl fumarate; N=number of subjects who received a specific treatment (subjects could be included in more than one treatment group); n=number of subjects included in the calculation of the geometric least squares means and SE, which is the number of subjects with specific PK data while on any of the dose regimens; PK=pharmacokinetic; SE=standard error.

[a] Based on ANCOVA, including treatment sequence, treatment group, and period as fixed factors and subject within treatment sequence as a random factor.

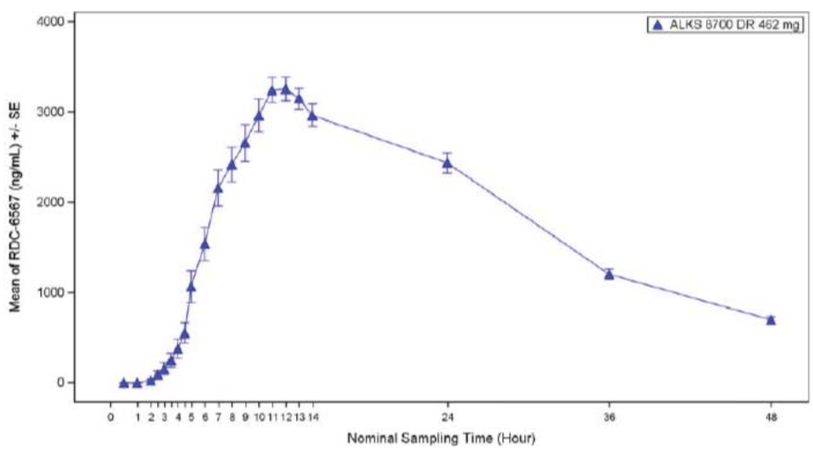
Note: The SE displayed is in logarithmic scale.

Source: Module 5.3.1.2 Study A104, Table 9.

MMF Plasma Concentration (Mean±SE) after Administration of DRF and DMF with a High-Fat and High-Calorie Meal (PK Population; Study A104)



HES Plasma Concentration (Mean±SE) after Administration of DRF and DMF with a High-Fat and High-Calorie Meal (PK Population; Study A104)



Study A109 was a Phase 1, single-center, open-label, randomized, four-period, four-sequence crossover study designed to evaluate the effect of fat and caloric content of a meal on the PK of MMF following a single dose of DRF or DMF relative to a single dose of DMF administered under fasted conditions.

Subjects were randomized in a 1:1:1:1 ratio to a sequence of four treatments:

•DMF 240 mg under fasted conditions (dosed after a minimum of 10 hours fasting [Group A])

•DMF 240 mg dosed with a high-fat/high-calorie meal (972 Kcal, 52.8% fat; Group B)

•DRF 462 mg dosed with a low-fat/low-calorie meal (406 Kcal,27.3% fat; Group C)

•DRF 462 mg dosed with a medium-fat/medium-calorie meal (698 Kcal, 38.0% fat; Group D)

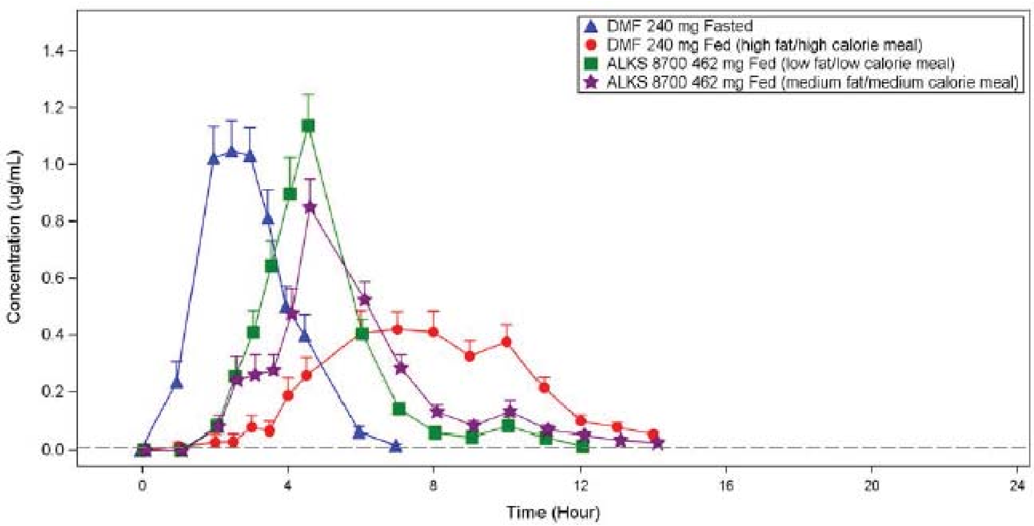
Forty-eight healthy subjects were randomized to receive study drug in one of the four sequences. Of the 48 subjects who received the study drug, a total of 46 completed the study and 2 subjects discontinued from the study early.

<u>Study A109 PK Summary:</u>

DRF plasma concentrations for all subjects were below the LLOQ following treatment with DRF.

The mean plasma concentration-time profiles of MMF were characterized by a delay in the fed conditions with the longest t_{lag} following administration of DMF with a high-fat/high-calorie meal. Following administration of DMF with a high-fat/high-calorie meal, MMF C_{max}, AUC_{last}, and AUC_{∞} decreased by 41%, 18%, and 3%, respectively compared to these after DMF administration in the fasted state.

MMF Plasma Concentration (Mean + SE) in Linear Scale (Study 109, PK Population)



Abbreviations: DMF=Dimethyl fumarate; MMF=Monomethyl fumarate; PK=pharmacokinetics; SE=standard error
Source: Figure 14.2.1.1

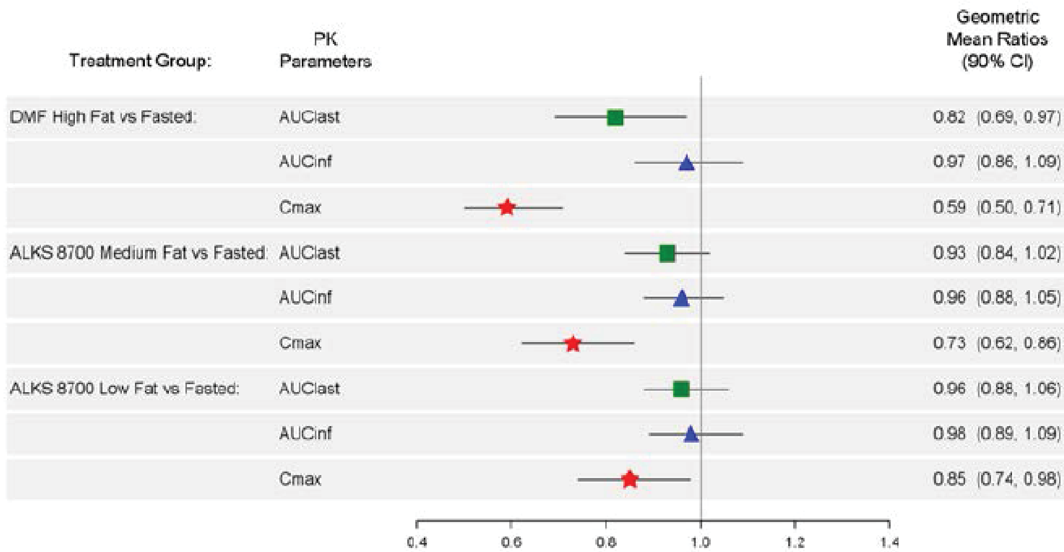
Meals of increasing fat and calories resulted in a decrease in MMF C_{max} with the most significant decrease (mean decrease of 41%) after administration of DMF following a high-fat/high-calorie meal, a mean 27% decrease after administration of DRF following a medium-fat/medium-calorie meal, and a mean decrease of 15% after administration of DRF following a low-fat/ low-calorie meal. The food effect on MMF AUC_{last} was less significant with reductions of 18%, 7%, and 4% for the DMF high-fat/high-calorie, DRF medium-fat/medium-calorie, and DRF low-fat/low-calorie treatments, respectively. The MMF AUC_{last} geometric mean ratios for the DRF treatments were close to 1 and the 90% CIs were contained within the 0.8 to 1.25 bounds.

Summary of Pharmacokinetic Parameters for MMF by Treatment Group (Study 109)

Parameter Statistics	Treatment Group			
	DMF 240 mg Fasted (N=48)	DMF 240 mg Fed (high fat/high calorie meal) (N=48)	ALKS 8700 462 mg Fed (low fat/low calorie meal) (N=48)	ALKS 8700 462 mg Fed (medium fat/medium calorie meal) (N=48)
AUC_{∞} (h•µg/mL)				
n	43	23	41	33
Arithmetic Mean (SD)	3.06 (0.98)	3.05 (1.04)	3.10 (0.92)	2.94 (0.75)
CV (%)	31.88	34.28	29.86	25.49
Median	2.98	3.00	3.22	2.88
Min - Max	1.6 - 6.0	1.1 - 5.4	0.8 - 5.4	1.2 - 4.6
Geometric Mean	2.92	2.86	2.94	2.84
AUC_{last} (h•µg/mL)				
n	48	47	47	47
Arithmetic Mean (SD)	3.05 (0.96)	2.78 (1.16)	2.96 (0.93)	2.83 (0.80)
CV (%)	31.45	41.92	31.30	28.37
Median	2.94	2.91	2.83	2.85
Min - Max	1.6 - 6.0	0.1 - 5.3	0.8 - 5.3	0.9 - 4.6
Geometric Mean	2.91	2.36	2.80	2.70
C_{max} (µg/mL)				
n	48	47	47	47
Arithmetic Mean (SD)	1.75 (0.70)	1.12 (0.54)	1.53 (0.69)	1.33 (0.63)
CV (%)	39.81	47.90	44.92	47.52
Median	1.67	1.07	1.46	1.29
Min - Max	0.6 - 3.5	0.1 - 2.4	0.2 - 3.5	0.2 - 3.0
Geometric Mean	1.62	0.95	1.38	1.17

A forest plot of geometric mean ratios comparing MMF PK parameters after administration of DMF following a high-fat/high-calorie meal, DRF following a medium-fat/ medium-calorie meal, or DRF following a low-fat/low-calorie meal relative to DMF administration in the fasted state is shown below.

Forest Plot of Geometric Mean Ratios of MMF Relative to DMF Administration in the Fasted State (PK Population; Study A109)



Abbreviations: ALKS 8700=diroximel fumarate; AUCinf (AUC_{∞})=area under the concentration-time curve from time zero extrapolated to infinity; AUClast (AUC_{last})=area under the concentration-time curve from time zero until the last measurable concentration time point; CI=confidence interval; Cmax (C_{max})=maximum plasma concentration; DMF=dimethyl fumarate; GMR=geometric mean ratio; MMF=monomethyl fumarate; PK=pharmacokinetic.

Note: Geometric mean and GMR were obtained from back transformation of the least squares mean and least squares mean difference in the log scale. The green squares represent the GMR of the AUC_{last} of the treatment group to the AUC_{last} of the DMF fasted group; the blue triangles represent the GMR of the AUC_{∞} of the treatment group to the AUC_{∞} of the DMF fasted group; the red stars represent the GMR of the C_{max} of the treatment group to the C_{max} of the DMF fasted group; and the bars on either side of each marker indicate the 90% CI of the GMR.

Source: Module 5.3.1.2 Study A109, Figure 8.

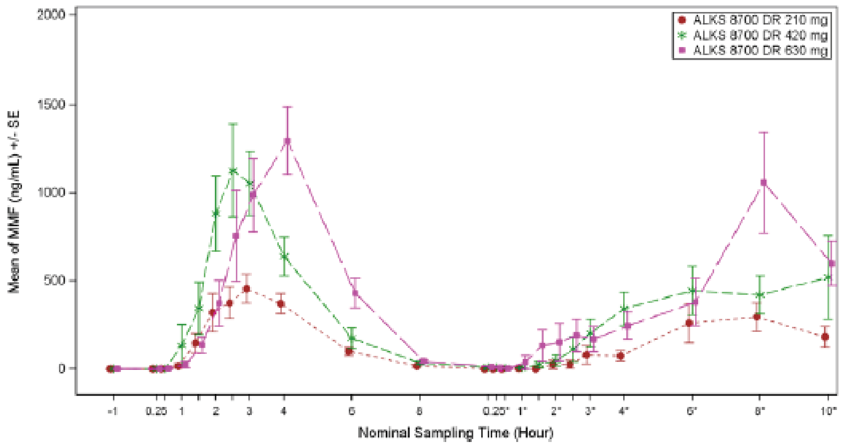
Discussion:

Based on these findings, subjects participating in the Phase 3 studies were instructed (b) (4) (b) (4) to avoid a high-fat, high-calorie meal; sponsor proposes same language in the product label.

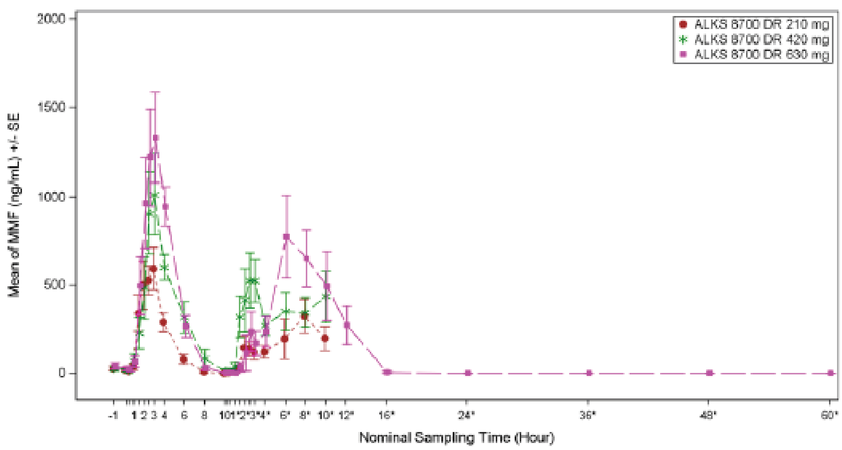
In study 102 Part B, on Day 1 and 5, three dose levels of DRF (210, 420, and 630 mg) were explored. The morning DRF dose was administered in the fasted state, the evening dose: fed (standardized meals). Following morning administration of 420 mg DRF (fasted), the mean C_{max} of MMF on Day 1 was 1.736 µg/mL; mean C_{max} of MMF on Day 5 was 1.455 µg/mL. Following the evening administration of 420 mg DRF in the fed state (standardized meals), the mean C_{max} of MMF on Day 1 was 1.136 µg/mL; mean C_{max} of MMF on Day 5 was 1.212 µg/mL. Assuming dose proportionality, these C_{max} values correspond to 1.25 and 1.33 µg/mL, respectively (if the DRF dose were 462 mg as proposed and as in all other clinical studies).

Study A102- Part B: MMF Plasma Concentration (Mean ±SE) for Days 1 and 5

Day 1



Day 5



*Note: *=PM dose*

Summaries of the MMF PK parameters from all clinical studies in the DRF development program, after DRF administration with or with food of various fat content and after co-ingestion with alcohol, are provided in the tables below.

MMF: Arithmetic Mean (SD) Cmax and AUC after a single-dose or multiple-dose administration of DRF 462 mg in the fasted state

Study	**Dose**	**Cmax µg/mL**	**AUC∞ hr*µg/mL**
A102 (Part A)	Single	1.63 (0.768)	3.09 (0.09)
A102 (Part B)	Single	1.74 (0.828)	3.697 (0.723)
A102 (Part B)	**Multiple**	1.46 (0.722)	NC
A103	Single	1.70 (0.699)	3.81 (1.16)

A105 (Trt A)	Single	1.49 (0.483)	3.29 (0.594)
A105 (Trt B)	Single	1.91	4.41 ()
A106	Single	1.36 (0.598)	3.06 (0.925)
A107	Single	1.62 (0.808)	3.32 (1.08)
A107	**Multiple**	1.62 (0.655)	3.56 (1.06)
A108	Single	1.45 (0.863)	3.34 (1.10)
A110	**Multiple**	1.49 (0.749)	4.01 (1.49)

MMF: Arithmetic Mean (SD) Cmax and AUC after a single-dose or multiple-dose administration of DRF at 462 mg/dose in the fed state

Study	**Food Type**	**Cmax** µg/mL	**AUC∞** hr*µg/mL
A102	Single/ High-fat	0.955 (0.630)	AUC_{last} 2.50 (0.10)
A102 (Part B) 420 mg DRF	Single/ std meal	1.136 (0.81)	NC
A102 (Part B) 420 mg DRF	Multiple/ std meal	1.212 (0.57)	NC
A104	Single/ High-fat	0.99 (0.507)	
A109	Single/ Low-Fat	1.53 (0.69)	3.10 (0.92)
A109	Single/ Medium-Fat	1.33 (0.63)	2.94 (0.75)
A109	High-Fat	0.99 (0.51)	2.96 (0.62)
A301	Multiple/ **Avoid high-fat**	1.31 (1.011)	

Reviewer's Comments:

In the phase 3 study A301 and in Phase 1 study 102 (Part B) patients/healthy subjects were instructed to take DRF with food (but avoid high-fat meal). In these studies, the mean C_{max} of MMF was similar (or higher) than MMF C_{max} after DMF administration with high fat meal. Following administration of DMF with a high-fat meal, the arithmetic mean MMF C_{max} was 1.12 µg/ml. In Study A104, MMF mean C_{max} value was 1.376 µg/ml after 240 mg DMF administered with high fat meal. After DRF with medium-fat meal mean MMF C_{max} was 1.33 µg/ml (see Study 109 Summary of Pharmacokinetic Parameters for MMF by Treatment Group above).

In addition, fed status had little effect on MMF AUC: DRF met the PK criteria for BE (to DMF) based on MMF AUC in three Phase 1 studies (Studies A103, A104, and A109).

Recommendations: Since taking the drug twice a day in the fasted state could be very challenging, and considering the above results, “Avoid high-fat, high-calorie meal” is acceptable as labeling recommendation. DNP clinical review team agreed with this recommendation.

3.3.5 Are there clinically relevant drug-drug interactions and what is the appropriate management strategy?

In vitro DDI studies

In vitro cytochrome P450 (CYP) inhibition and induction studies and P-gp studies conducted with Tecfidera indicated no potential for drug interactions with DMF or MMF (Tecfidera USPI). Diroximel fumarate is hydrolyzed by esterases to produce MMF, RDC-8439, and HES as demonstrated in an in vitro study using human liver microsomes and intestinal S9 fractions (Study XT134113).

Diroximel fumarate, HES and RDC-8439 did not inhibit CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6, and CYP3A4/5 enzymes in human liver microsomes at concentrations up to 200 µM (51.0 µg/mL) (Studies XT135135 and XT13A014).

No induction of CYP1A2, CYP2B6, or CYP3A4 activity or of CYP mRNA was observed in human hepatocytes by DRF (up to 300 µM), HES (up to 5000 µM), and RDC-8439 (up to 300 µM) (Study XT153079). There was a 2.34-fold change at 5000 µM HES in CYP2B6 and 2.12-fold change in CYP3A4 mRNA levels in one of the three cultures tested; however, this should not be clinically relevant as steady-state Cmax levels in humans are about 76 µM.

Diroximel fumarate inhibited P-gp mediated transport by 41% at high dose (100 µM), while RDC-8439 and HES did not inhibit P-gp mediated transport at the concentrations tested (10 and 100 µM; Study XT138097). Diroximel fumarate and its metabolites did not induce P-gp (Study 17ALKRP1R2).

The results of another in vitro study (b) (4) Protocol No: Alkermes-01-09Jul2017) indicated that the potential for clinical drug-drug interaction with DRF, HES, and RDC-8439 due to transporter interactions is low. HES and DRF were not substrates of breast cancer resistance protein (BCRP) in vitro. Diroximel fumarate and HES were not inhibitors of BCRP. RDC-8439 was likely a substrate of BCRP; however, due to its low permeability and high variability in net efflux ratio, these results are expected to have minimal clinical consequence. In addition, RDC-8439 is a minor metabolite. HES was neither a substrate nor an inhibitor of multidrug and toxin extrusion transporter (MATE)1, MATE2-K, organic anion transporter (OAT)1, OAT3 or organic cation transporter (OCT)2.

Alcohol DDI

In vitro dissolution studies have shown that DRF (ALKS 8700), delayed release capsule (231 mg), demonstrates alcohol dose dumping in the order of >50% in 20% ethanoic acid. Therefore, a clinical study was conducted to determine the influence of co-ingestion of alcohol on the PK of DRF and its metabolites in healthy subjects.

Study A106 was a Phase 1, single-center, open-label, randomized, three-period, six-sequence, crossover study in healthy adult subjects to evaluate the potential interaction between alcohol consumption and diroximel fumarate PK.
A total of 31 subjects were randomized, 5 subjects per treatment sequence.

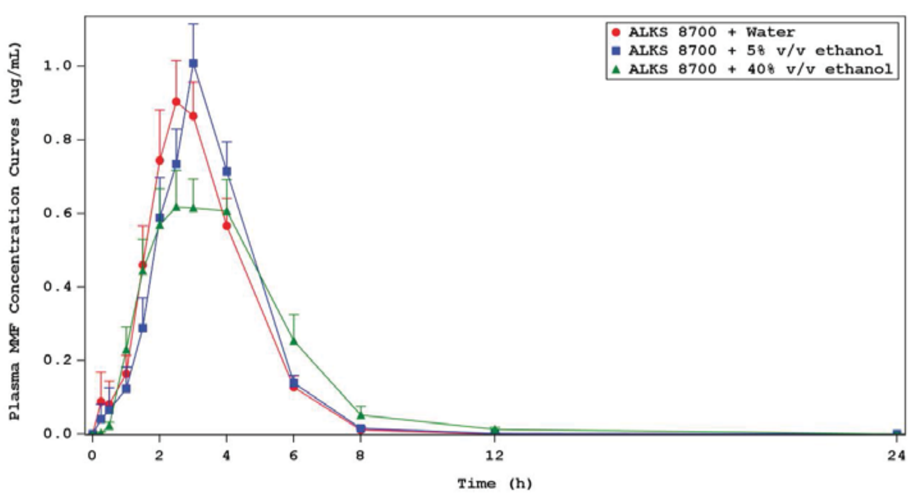
Treatment Sequence	Treatment Period 1	Treatment Period 2	Treatment Period 3
Sequence 1	A	B	C
Sequence 2	B	C	A
Sequence 3	C	A	B
Sequence 4	C	B	A
Sequence 5	A	C	B
Sequence 6	B	A	C

Abbreviations: A=distilled water, B=5% ethanol–distilled water solution, C=40% ethanol solution

On Days 1, 8, and 15, a single oral dose of ALKS 8700 462 mg (two capsules of 231 mg) was administered at approximately the same time each morning with 240 mL of either distilled water, 5% v/v ethanol–distilled water solution, or 40% v/v ethanol solution, as per sequence. Subjects were required to fast for a minimum of 10 hours prior to dosing and until 4 hours post-dose.

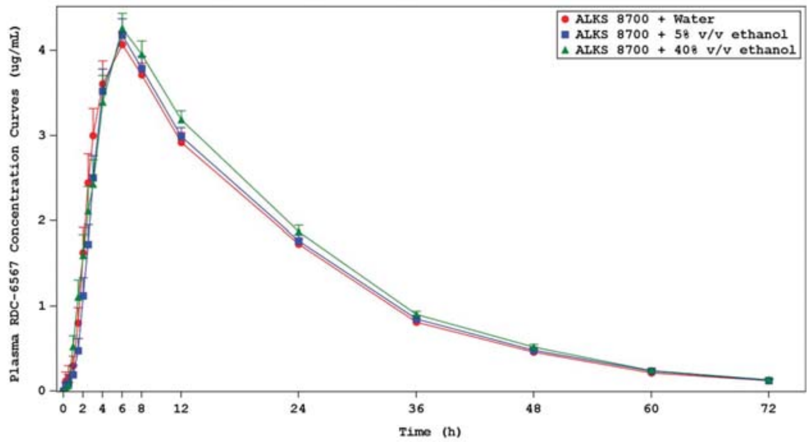
Mean plasma MMF concentrations were comparable for ALKS 8700 + water and ALKS 8700 + 5% v/v ethanol. After reaching similar peak plasma levels (approximately 0.9−1.0 µg/mL), the MMF concentrations declined at similar rates for these two treatments. However, the mean peak plasma MMF concentration for ALKS 8700 + 40% v/v ethanol was lower (approximately 0.6 µg/mL) and declined slower.

Study A106: Mean (+SE) Plasma MMF Concentration Curves



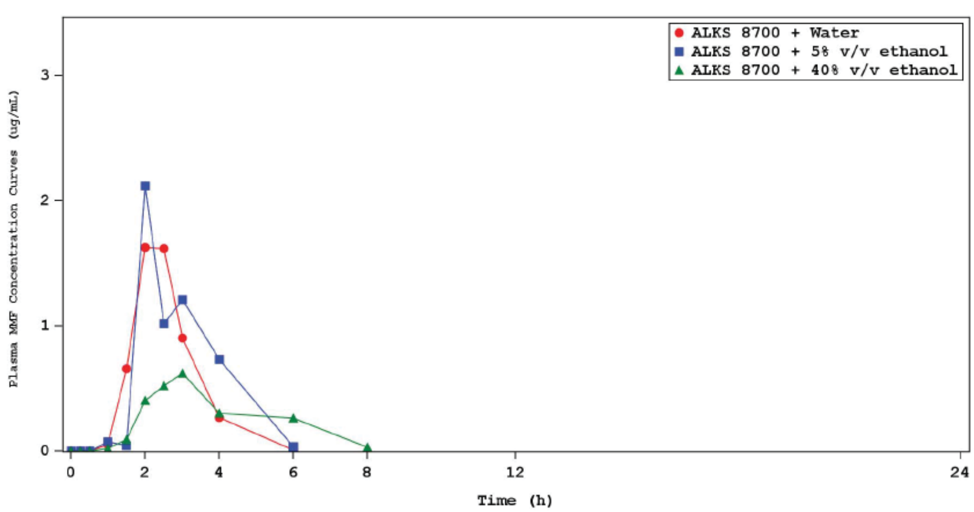
HES PK was unaffected by the co-ingestion of ethanol with DRF.

Study A106: Mean (+SE) Plasma HES (RDC-6567) Concentration Curves



Comments: There was a significant (more than 50%) decrease in C_{max} with 40% alcohol in 10 out of 31 subjects. Two subjects had missing MMF concentrations after 40% alcohol. Of the 31 subjects who received at least one dose of study drug, 30 completed the study (ie, completed the three treatment periods and the Follow-up visit). One subject discontinued early. One subject ((b) (6) experienced vomiting 1 hour post co-ingestion of 40% ethanol with DRF. The median Tmax of MMF is approximately 3 hours. Because the incidence of vomiting was within the first 6 hours (2× the median Tmax), the MMF concentration data of that subject was excluded from the overall non-compartmental analysis.

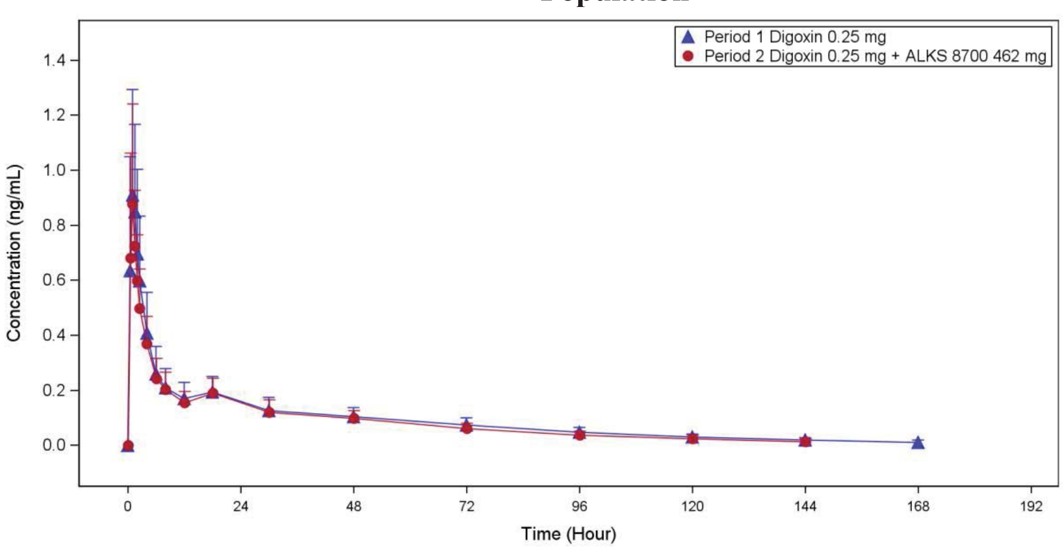
Individual Plasma MMF Concentration Curves (ug/mL)



Recommendations: Administration of Vumerity with alcohol should be avoided.

Digoxin DDI study (ALK8700-A107)
Based on in vitro studies, DRF has been found to inhibit P-gp mediated transport of digoxin by 41% at the highest concentration tested (100 μM). Therefore, due to the presence of DRF in the intestine, where it is metabolized primarily to MMF and RDC-6567 and based on the FDA Guidance for Drug Interaction Studies (2012), a drug interaction clinical study was conducted to investigate the effect of DRF on the PK of digoxin in healthy adult subjects.
24 healthy adult subjects were planned to be enrolled into this study.
On Day 1, digoxin 0.25 mg was administered orally. This was followed by a 10-day washout period between doses. Blood samples for digoxin PK analysis were obtained prior to dosing on Day 1 and through to 192 hours post-digoxin dosing. On the morning of Day 11, 462 mg DRF (given orally as two capsules of 231 mg) was administered. Approximately 30 minutes after DRF dosing, digoxin 0.25 mg was administered orally. On the evening of Day 11, the second dose of DRF 462 mg was administered. On Days 12 to 15, DRF 462 mg was administered twice daily (to maintain DRF exposure over the digoxin concentration-time course).
Blood samples for digoxin PK analysis were obtained prior to dosing on Day 11 and through to 192 hours post-digoxin dosing, when the last digoxin sample was collected on Day 19. Separate blood samples for PK analysis of DRF were obtained prior to dosing on Days 11 and 15, and through 12 hours post-dose.
Digoxin (on Days 1 and 11) and the morning dose of DRF (on Days 11 and 15) were administered after an overnight fast of least 10 hours. No food was allowed for at least 4 hours after dosing. All other DRF doses were administered without regard to restrictions for food or water; however, subjects received standardized meals scheduled at the same time throughout the inpatient period of the study.
Results: The mean concentration-time profiles for digoxin when digoxin was administered with DRF compared to when it was administered alone are shown below.

Plasma Concentration Time Profile of Digoxin in the Presence and Absence of DRF – PK Population



Source: Figure 14.2.1.1

Summary of PK Parameters for Digoxin in the Presence and Absence of DRF

PK Parameter (Unit) Statistics	Period 1 Digoxin 0.25 mg (N=24)	Period 2 Digoxin 0.25 mg + ALKS 8700 462 mg (N=23)
AUC_{∞} (h•ng/mL)		
n	23	23
Arithmetic Mean (SD)	16.090 (5.023)	13.922 (4.146)
CV (%)	31.216	29.777
C_{max} (ng/mL)		
n	24	23
Arithmetic Mean (SD)	1.009 (0.358)	0.918 (0.349)
CV (%)	35.497	38.018
t½ (h)		
n	24	23
Arithmetic Mean (SD)	43.290 (10.633)	34.701 (4.499)
CV (%)	24.563	12.966
Median	41.716	34.053

PK Summary:
Exposure of digoxin, as measured by C_{max}, AUC_{last}, and AUC_{∞}, decreased approximately 10% in the presence of DRF. However, the similarity of the profiles over the first 72 hours suggests that DRF does not inhibit P-gp in vivo.
Following single and multiple administrations of DRF in the presence of digoxin, exposure to MMF, RDC-6567, and RDC-8439, as measured by Cmax and AUC, was consistent with the known PK profile of DRF.

Recommendations: No labeling recommendations are needed as Vumerity does not inhibit P-gp in vivo.

4. APPENDICES

4.1: Summary of Bioanalytical Method Validation and Performance

4.1.1 How are the active moieties identified and measured in the clinical pharmacology and biopharmaceutics studies?

Concentrations of DRF and its metabolites MMF, HES, and RDC-8439 were measured by validated bioanalytical methods in plasma. Two liquid chromatography with tandem mass spectrometry (LC-MS/MS) detection bioanalytical methods were developed and validated by (b) (4) ((b) (4) 14-RDC5108-01 and (b) (4) 15027). The method for determination of HES in plasma was developed after Study 001, so samples from Part 1 of Study 001 were analyzed for HES measurement in a subsequent addendum.

In addition, MMF and HES were measured by validated bioanalytical methods in urine.

(b) (4) 14-RDC5108-01):

An assay measuring DRF, MMF, and RDC-8439 concentrations in human dipotassium ethylenediaminetetraacetic acid (K2-EDTA) plasma containing 10 mM citric acid and 1% formic acid was developed and validated at (b) (4) 14-RDC5108-01). The method used internal standards (b) (4) (for DRF) and (b) (4) (for RDC-8439 and MMF).

The calibration curve for each analyte ranged from 25.0 to 10000 ng/mL.

QC samples were prepared at six concentrations (25.0, 75.0, 500, 8000, 10000, and 20000 ng/mL).

Dilution integrity was verified at a concentration up to 20000 ng/mL when diluted 50-fold.

The validation results are summarized below.

(b) (4) 14-RDC5108-01 Validation Summary

Assay Range	25.0 to 10000 ng/mL				
Regression Model and Linearity	Weighted linear (1/concentration^2)				
Selectivity (Specificity)	No significant interference at the retention time of diroximel fumarate (b) (4) RDC-8439, MMF, or (b) (4) was observed in any of the human plasma lots screened				
Precision and Accuracy	**QC**	**Precision (%CV)**		**Accuracy (%RE)**	
		Within-run	**Between-run**	**Within-run**	**Between-run**
	Diroximel Fumarate[a]				
	LLOQ 25.0 ng/mL	4.6	5.0	20.0	19.2
	LQC 75.0 ng/mL	8.4	8.9	6.8	-3.6
	MQC 500 ng/mL	4.0	7.7	-1.4	-3.6
	HQC 8000 ng/mL	3.6	7.1	-2.1	-1.8
	ULOQ 10000 ng/mL	3.8	9.6	-10.3	-0.2
	DQC 20000 ng/mL	3.6	NR	-4.0	NR
	MMF[b]				
	LLOQ 25.0 ng/mL	4.5	5.6	18.8	16.4
	LQC 75.0 ng/mL	6.8	8.5	-4.4	-6.9
	MQC 500 ng/mL	4.9	7.8	-10.2	-7.2
	HQC 8000 ng/mL	4.5	6.2	3.8	3.1
	ULOQ 10000 ng/mL	5.3	6.3	-3.6	3.0
	DQC 20000 ng/mL	5.5	NR	11.5	NR

Precision and Accuracy	**RDC-8439**[c]				
	LLOQ 25.0 ng/mL	5.2	4.9	20.0	18.8
	LQC 75.0 ng/mL	3.8	7.7	0.7	-3.1
	MQC 500 ng/mL	4.5	7.5	1.4	-5.4
	HQC 8000 ng/mL	5.3	8.3	-3.1	-2.8
	ULOQ 10000 ng/mL	3.4	7.9	-4.4	-5.6
	DQC 20000 ng/mL	4.2	NR	-7.5	NR
Short-term Stability	6 hours on benchtop on ice				
Long-term Stability	Diroximel fumarate: 32 days at -70°C MMF: 19 days at -70°C RDC-8439: 20 days at -70°C				
Reinjection Integrity	26.0 hours at 10°C				
Processed Sample Stability	26.3 hours at 10°C				
Freeze-Thaw Stability	Diroximel fumarate: 3 cycles at -70°C MMF and RDC-8439: 4 cycles at -70°C for LQC, MQC, and HQC samples, 3 cycles at -70°C for DQC				
Solution Stability[d]	7.2 hours at room temperature; 23 days at -70°C				
Interference	No interference between the 3 analytes				

Source: Study (b) (4) 14-RDC5108-01.

(b) (4) 15027):

An assay for DRF, MMF, and RDC-8439 in human K2-EDTA plasma acidified with 1% citric acid (10 mM final concentration) and 1% formic acid was developed and validated at (b) (4) (b) (4)

The method used internal standards (b) (4) (for DRF) and (b) (4) (for RDC-8439 and MMF).

The calibration curve for each analyte ranged from 10.0 to 4000 ng/mL.

QC samples were prepared at three concentrations (30.0, 200, and 3200 ng/mL).

Dilution integrity was verified at a concentration up to 16000 ng/mL when diluted 20-fold.

The validation results are summarized below.

(b) (4) 15027 Validation Summary

Assay Range	10.0 to 4000 ng/mL
Regression Model and Linearity	Weighted linear (1/concentration^2)
Selectivity (Specificity)	No significant interference at the retention time of diroximel fumarate, (b) (4) RDC-8439, MMF, or (b) (4) was observed in any of the six human plasma lots screened

Precision and Accuracy[a]	QC [QC Study][b]	Precision (%CV)		Accuracy (%RE)	
		Within-run	Between-run	Within-run	Between-run
	Diroximel Fumarate				
	LLOQ 10.0 ng/mL	4.27 to 10.04	7.65	-8.30 to -2.80	-4.60
	LQC 30.0 ng/mL	3.72 to 9.35 [2.35]	6.45	2.33 to 6.33 [7.67]	4.33
	MQC 200 ng/mL	5.41 to 7.77 [3.02]	7.48	1.00 to 11.00 [7.50]	7.00
	HQC 3200 ng/mL	3.10 to 5.65 [1.89]	8.76	-13.75 to 2.50 [7.19]	-4.06
	DQC 16000 ng/mL	5.77	NR	1.88	NR
	MMF				
	LLOQ 10.0 ng/mL	7.90 to 11.83	9.35	4.00 to 11.00	7.00
	LQC 30.0 ng/mL	2.23 to 8.57 [3.73]	7.35	-1.67 to 7.00 [3.67]	3.33
	MQC 200 ng/mL	1.77 to 3.60 [1.03]	6.21	-6.00 to 7.00 [4.00]	1.50
	HQC 3200 ng/mL	2.00 to 2.81 [1.46]	7.21	-7.81 to 8.13 [1.56]	-0.31
	DQC 16000 ng/mL	1.99	NR	6.25	NR
	RDC-8439				
	LLOQ 10.0 ng/mL	2.28 to 5.97	5.22	-0.60 to 5.00	3.00
	LQC 30.0 ng/mL	3.33 to 9.11 [2.45]	7.60	-4.67 to 5.00 [6.33]	1.33
	MQC 200 ng/mL	3.13 to 5.28 [3.99]	6.60	-6.00 to 7.00 [9.50]	1.50
	HQC 3200 ng/mL	2.45 to 6.11 [2.55]	8.13	-8.13 to 8.44 [0.00]	0.31
	DQC 16000 ng/mL	3.44	NR	7.50	NR

Short-term Stability	5 hours on benchtop at approximately 4°C
Long-term Stability	Diroximel fumarate: 31 days at approximately -20°C, 80 days at approximately -70°C MMF and RDC-8439: 80 days at approximately -20°C, 80 days at approximately -70°C
Reinjection Integrity	Diroximel fumarate: 4 hours at room temperature; 191 hours at approximately 4°C MMF and RDC-8439: 143 hours at room temperature; 191 hours at approximately 4°C
Processed Sample Stability	Diroximel fumarate: did not meet acceptance criteria[c] MMF: 72 hours at room temperature; 106 hours at approximately 4°C RDC-8439: 78 hours at room temperature; 99 hours at approximately 4°C
Freeze-Thaw Stability	5 cycles at approximately -70°C
Solution Stability	Diroximel fumarate and RDC-8439: 24 hours at room temperature, 207 days at approximately -20°C MMF: 24 hours at room temperature, 209 days at approximately -20°C Internal standard (b) (4) 18 hours at room temperature, 206 days at approximately -20°C Internal standard (b) (4) 18 hours at room temperature, 101 days at approximately -20°C
Extraction Recovery	Diroximel fumarate: 79.2% to 81.2% MMF: 80.1% to 81.2% RDC-8439: 78.8% to 80.7% (b) (4) 83.9% to 87.3% (b) (4) 91.4% to 93.7%
Interference	No interference from HES or digoxin

Source: Study (b) (4) 15027 (Report: RPT03973)

Information Request for clarification was sent on 29 Jan 2019:

We note that several bioanalytical methods for analysis of MMF in plasma samples (validated by two bioanalytical laboratories (b) (4)) are listed in Modules 2.7.1 and Module 5.3.1.4. Please clarify whether these methods were cross-validated.

Sponsor responses (Response to Agency RFI_29 Jan 2019).

All bioanalytical methods used to support the Phase 1 and Phase 3 studies were fully validated independently. Cross-validation was not included with the application because the analysis of all pharmacokinetic samples was conducted using only one method at one bioanalytical laboratory per study (Table 1). Neither multiple methods nor bioanalytical laboratories were used within a single study. Each biopharmaceutical study was designed with inclusion of the appropriate reference treatment arms to draw the conclusions. Furthermore, analysis of the maximum concentration (Cmax) data obtained across studies demonstrates the consistency of the methods with respect to MMF concentration. There was a substantial overlap of the Cmax values of MMF at a given dose when compared between two studies, indicating that there is no significant difference in values obtained via two different methods. A summary of the MMF Cmax values from studies conducted with similar dose administration conditions is provided in Table 2. Therefore, the full validations of the bioanalytical methods ensure the reliability of the data.

Table 1: Summary of Studies by Bioanalytical Method

Analytical Laboratory/ Validation Report No.	(b) (4)		
	(b) (4) 14-RDC5108-01	(b) (4) 15027	(b) (4) 15125
Matrix	Acidified Plasma	Acidified Plasma	Non-Acidified Plasma
Studies	001	A105	A110
	A102	A106	A301
	A103	A107	A302
	A104	A108	--
	--	A109	--

Table 2: Summary of Cmax [Mean (SD)] Values by Bioanalytical Method

Analytical Laboratory	(b) (4)					
Matrix	Acidified Plasma		Acidified Plasma		Non-Acidified Plasma	
Dose	Study	C_{max} (mg/mL)	Study	C_{max} (mg/mL)	Study	C_{max} (mg/mL)
462 mg Fasted, Single Dose	A103	1.70 (0.699)	A105	1.49 (0.483)	--	--
	--	--	A106	1.36 (0.598)	--	--
	--	--	A107	1.62 (0.808)	--	--
	--	--	A108	1.45 (0.863)	--	--
DMF Fasted, Single Dose	A103	1.83 (0.760)	A109	1.75 (0.70)	--	--
462 mg Fasted, Multiple Dose	--	--	A107	1.62 (0.655)	A110	1.49 (0.749)

[1] Only studies under fasted conditions were included for comparison given the known effect of food on C_{max} values with higher interindividual variability.

Additional Comments:

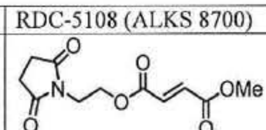
The following methods were used for analysis of MMF:

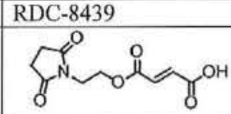
MMF Plasma Validation Study (b) (4) 15125

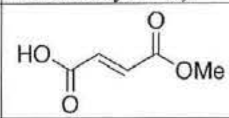
MMF Acidified Plasma Validation Study (b) (4) 15027

MMF Acidified Plasma Validation Study (b) (4) 4-RDC5108-01

Additional concerns: internal standards (b) (4) for DRF (ALKS 8700) and (b) (4) (for RDC-8439). Since the highlighted molecules have quite different structures and MW (and behave differently during the sample preparation and the LC-MS/MS analysis), sponsor needs to justify using them as IS. This would be clarified with the lab during OSIS inspection.

Chemical Name	RDC-5108 (ALKS 8700)
Chemical Structure	
Formula	$C_{11}H_{13}NO_6$
Molecular weight (MW)	255.22
Monoisotopic MW	255.07

Chemical Name	RDC-8439
Chemical Structure	
Formula	$C_{10}H_{11}NO_6$
Molecular weight (MW)	241.20
Monoisotopic MW	241.06

Chemical Name	MMF (monomethyl fumarate or fumaric acid monomethyl ester)
Chemical Structure	
Formula	$C_5H_6O_4$
Molecular weight (MW)	130.10
Monoisotopic MW	130.03

2.1.2 Internal Standard

(b) (4)

OSIS Inspections

Inspection Request was issued for the pivotal studies in this clinical development program, e.g. Studies ALK8700-A103, A104 and A109.

The clinical site, PPD, Inc. and one of the bioanalytical sites, (b) (4) have been recently inspected and no issues were identified, therefore OSIS declined to inspect these sites.

The Decline to Inspect Memo were uploaded in DARRTS on March 13th, for PPD for (b) (4) (b) (4).

OSIS conducted an inspection of (b) (4) and concluded that the data from the audited studies are reliable to support a regulatory decision (Report issued on July 26, 2019).

During inspection, OSIS asked the firm to justify the use of (b) (4) and (b) (4) as internal standards (IS) for RDC-5108 and RDC-8439, respectively. They stated that the IS tracked their respective analytes as demonstrated by the acceptable analytical performance from the global accuracy and precision data for RDC-5108 and RDC-8439 in Study ALK8700-A109, causing no concern regarding their appropriateness in quantitation of the analytes.

OSIS Evaluation: The firm's response did not address the question whether the selected ISs track the respective analytes. However, during the inspection, OSIS did not find any indication that the choice of the IS for RDC-5108 and RDC-8439 adversely affected data reliability for Study ALK8700-A109.

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

HRISTINA DIMOVA
08/30/2019 04:45:32 PM

YUXIN MEN
09/03/2019 10:56:18 AM